

Research Proposal

Has the Wealth-Fertility Relationship Really Inverted? A Cross-Cohort, Cross-Country Test of the Demographic Transition Fitness Paradox

MSc Applied Social Data Science, Trinity College Dublin

1. Summary

The demographic transition fitness paradox - the observation that higher-status individuals in modern populations have fewer children, contradicting the central prediction of behavioural ecology, remains the most prominent anomaly in human evolutionary demography. This dissertation tests four specific hypotheses about the shape, timing, and mechanisms of the socioeconomic-fertility gradient across European cohorts born 1930 – 1980. Using education-stratified fertility data from the Wittgenstein Centre, attitudinal data from the European Social Survey, country-level covariates from V-Dem and the OECD, and individual-level cohort data from the 1970 British Cohort Study, the project combines panel econometrics, quantile and semi-parametric regression, interpretable machine learning, and Bayesian hierarchical modelling. The contribution is threefold: a clean cross-cohort test of the non-monotonic (J-curve) hypothesis with modern data, a high-dimensional horse-race between cultural and economic explanations at country level, and a mediation decomposition of the education-fertility relationship at individual level.

2. Background and Motivation

In the standard model of behavioural ecology, organisms with more resources should produce more surviving offspring. For most of human history this prediction was upheld: wealth, status, and reproductive success co-varied positively across pre-industrial agrarian and forager populations. Beginning in the eighteenth and nineteenth centuries, the demographic transition decoupled these variables across most of Europe and subsequently the rest of the world. In contemporary high-income societies the relationship has often inverted: those with the highest education and income have, on average, the fewest children. Vining (1986) named this the "central theoretical problem in human sociobiology," and despite four decades of subsequent work the underlying mechanisms remain contested. Three families of explanation compete: cultural transmission of low-fertility norms from elites downward (Colleran; Bongaarts and Watkins); quantity-quality substitution under modern returns to human capital (Kaplan; Lawson and Mace); and intergenerational fitness over longer time horizons in which low-fertility, high-investment strategies still maximise descendants several generations out (Goodman, Koupil and Lawson).

Recent work has begun to question whether the relationship is in fact uniformly negative. Stulp and Barrett, Hopcroft, and others document a possible "J-curve" in which fertility ceases to fall, or rises again, at the very top of the income distribution. Whether this revival is real, where it occurs first, and what mechanisms drive it are open questions of direct relevance to evolutionary theory and to demographic forecasting.

3. Research Questions and Hypotheses

The central question is: across European cohorts born 1930–1980, what is the shape of the socioeconomic-fertility gradient by cohort and country, and what conditions predict its steepness and the timing of its inversion?

This decomposes into four testable hypotheses.

H1 (Inversion). For cohorts born 1930–1980 across studied European countries, the relationship between education and completed fertility is negative.

H2 (J-curve). For the most recent cohorts in the sample, the SES-fertility relationship is nonmonotonic, with fertility rising at the top of the educational or income distribution.

H3 (Cultural transmission). Cross-country variation in the timing and depth of inversion is predicted better by indicators of secularisation, individualism, and gender-norm change than by economic indicators (GDP per capita, female labour-force participation, contraceptive prevalence).

H4 (Individual mechanism). Within individuals in the 1970 British Cohort Study, the effect of education on completed fertility is mediated primarily by attitudinal and preference variables rather than by realised earnings.

The four hypotheses together cover descriptive pattern, distributional shape, country-level mechanism, and individual-level mechanism.

4. Theoretical Framework

The dissertation engages three theoretical traditions. From classical demographic transition theory it adopts the period-cohort distinction and the formal apparatus of education-stratified fertility. From human behavioural ecology it adopts the predictive framework of resourcefertility correlations and the quantity-quality tradeoff (Kaplan 1996; Borgerhoff Mulder 1998, 2009). From cultural evolutionary theory it adopts the model of fertility norms as socially transmitted traits subject to prestige bias and conformity (Colleran 2016; Mace 2008). The empirical strategy is designed to discriminate between these frameworks rather than to assume any one. The dissertation does not aim to resolve the paradox theoretically; it aims to produce sharp empirical results that constrain which explanations remain viable.

5. Data

Five datasets form the empirical spine.

Wittgenstein Centre Data Explorer. Education-stratified completed cohort fertility for European countries, by birth cohort and education level. This is the primary outcome data for H1, H2, and H3. Open access.

Human Fertility Database and Human Mortality Database. Period and cohort fertility and mortality rates for context, robustness, and decomposition. Open access.

European Social Survey (ESS), rounds 1–11. Individual-level attitudinal data on religiosity, gender norms, family values, and fertility intentions, aggregated to country-cohort cells for H3. Open access with registration.

V-Dem, OECD, World Bank, Eurostat. Country-year covariates including secularisation indices, female labour-force participation, GDP per capita, gender-equality indicators, welfare state generosity, and contraceptive prevalence. Open access.

1970 British Cohort Study (BCS70). Individual-level longitudinal data on a single birth cohort, including education, income, attitudes, partner characteristics, and completed fertility. Used for H4. Accessed via the UK Data Service with project registration.

A robustness sample using the 1958 National Child Development Study (NCDS) and, if time permits, the Wisconsin Longitudinal Study will be considered.

6. Methodology

Each hypothesis is tested with a method appropriate to the question and to the structure of the data.

H1. Fixed-effects panel regression of completed cohort fertility on education category, with country and cohort fixed effects pooled across the Wittgenstein data. The objective is to establish the descriptive fact carefully, identify country-cohort cells that deviate from the pooled pattern, and provide a baseline for subsequent analyses.

H2. Quantile regression and generalised additive models on the education-fertility relationship within cohorts. The J-curve is operationalised as a statistically detectable upward inflection in the upper tail of the SES distribution. Sensitivity to model specification is reported.

H3. A two-stage analysis. Stage one uses gradient-boosted regression trees (XGBoost) with SHAP values to identify which country-level features predict inversion timing and gradient steepness in a flexible non-parametric framework. Stage two estimates a Bayesian hierarchical regression with regularising priors on a parsimonious feature set selected on

theoretical grounds informed by stage one. The two-stage design separates exploratory variable discovery from confirmatory uncertainty quantification.

H4. Causal mediation analysis on the BCS70 sample using the double machine learning framework (Chernozhukov et al. 2018) to control for high-dimensional pre-treatment confounding. The total effect of education on completed fertility is decomposed into direct and indirect components, with mediation through attitudinal, earnings, and partnercharacteristic pathways. The 1972 raising of the school-leaving age in the United Kingdom provides a potential instrument for education in robustness checks.

All analyses will be conducted in R, using `tidyverse`, `fixest`, `mgcv`, `xgboost`, `iml`, `brms`, and `DoubleML` packages. The project will be implemented as a reproducible pipeline using `targets`, with code and pre-registered analysis plan deposited on the Open Science Framework prior to data analysis.

7. Expected Contributions

The dissertation aims to make three contributions. First, a methodologically modern crosscohort test of the J-curve hypothesis, which has been claimed in fragmentary form in several studies but not systematically tested across European cohorts using education-stratified data. Second, a principled high-dimensional comparison of cultural and economic explanations of cross-country variation, going beyond the small-N regressions typical in the literature. Third, an individual-level mediation analysis of the education-fertility relationship using double machine learning, which to the candidate's knowledge has not been published for the BCS70 cohort despite the data's availability.

A secondary contribution is methodological: demonstrating responsible integration of interpretable machine learning with classical demographic and causal-inference methods in a substantively-driven dissertation.

9. Ethical Considerations

All datasets used are either fully open-access or made available for academic research under standard data-access agreements. No new data are collected from human participants. BCS70 access will be obtained through the UK Data Service with appropriate project registration and secure data handling. No identifiable individuals will be analysed or reported. The preregistration of analyses on the Open Science Framework supports transparency and limits the risk of selective reporting. Ethical approval will be sought via the standard TCD process if required for secondary data analysis.

11. Selected References

Bongaarts, J. and Watkins, S. C. (1996). Social interactions and contemporary fertility transitions. *Population and Development Review*.

Borgerhoff Mulder, M. (1998). The demographic transition: Are we any closer to an evolutionary explanation? *Trends in Ecology and Evolution*.

Chernozhukov, V. et al. (2018). Double/debiased machine learning for treatment and structural parameters. *Econometrics Journal*.

Colleran, H. (2016). The cultural evolution of fertility decline. *Philosophical Transactions of the Royal Society B*.

Goodman, A., Koupil, I. and Lawson, D. W. (2012). Low fertility increases descendant socioeconomic position but reduces long-term fitness in a modern post-industrial society. *Proceedings of the Royal Society B*.

Kaplan, H. (1996). A theory of fertility and parental investment in traditional and modern human societies. *Yearbook of Physical Anthropology*.

Lawson, D. W. and Mace, R. (2011). Parental investment and the optimization of human family size. *Philosophical Transactions of the Royal Society B*.

Mace, R. (2008). Reproducing in cities. *Science*.

Stulp, G. and Barrett, L. (2016). Wealth, fertility and adaptive behaviour in industrial populations. *Philosophical Transactions of the Royal Society B*.

Vining, D. R. (1986). Social versus reproductive success: The central theoretical problem of human sociobiology. *Behavioral and Brain Sciences*.